

SDA Group Submission Assignment 5

For your group submission, upload one .pdf in Canvas 😊

If you follow all installation instructions given in Canvas and still have big issues with R Markdown installation, you can exceptionally submit a .pdf generated in a different way (e.g., by copying code ‘manually’ into a text editor), which should still include all relevant explanations, code snippets and graphics.

Always remember that there are several ways to code something, and that the useful R functions provided for each task are just some suggestions. Finally, the symbol ⚡ indicates something generally worthy of attention. Have fun!

Exercise 1

A large class takes a statistics exam and their (*fictitious*) grades are collected in file `grades.RDS` on Canvas. In particular, the R object saved in that file is a list of two elements (which can be extracted ‘by name’ using the operator `$`): vector `on_time`, containing the grades of the 250 student who started the exam on time, and vector `late`, containing the grades of the 25 who started the exam late due to traffic on their route to uni.

- Investigate the normality of each of the two samples by using the Shapiro-Wilk test with significance level $\alpha = 5\%$. Comment of the result of each test: is the null hypothesis rejected or not? Is a one-sample *t*-test an appropriate location test for this data and why/why not?
- Test the null $H_0 : m \leq 7$ against the alternative $H_1 : m > 7$ for the median m of the `on_time` sample using a sign test. Is the null rejected (use a significance level of $\alpha = 5\%$)?

Recall that the test statistic has null distribution $\text{Bin}(n, 0.5)$ (that is, binomial with parameters n – where n is the size of the sample – and $p = 1/2$) and that, for large n , the binomial distribution can be approximated with a normal distribution with mean np and standard deviation $\sqrt{n \cdot p \cdot (1 - p)}$. Compute the *p*-value of the preceding (right-sided) test using this normal approximation. How does it compare to the ‘true’ *p*-value and does it lead to the same test decision?

Hint: recall to check if any values in the sample are equal to $m_0 = 7$ before performing the sign test and, if necessary, adapt the testing procedure accordingly.

- Repeat the analysis you performed in b. for the `late` sample. How does the normal approximation perform in this case and why?

Deliverables

- For all subtasks, to aid in correction: all relevant code
- The test results and comments required by a.
- The test results, computations and comments required by b.
- The test results, computations and comments required by c.

Useful R functions

`readRDS`, `$`, `shapiro.wilk`, `any`, `sum`, `binom.test`, `pnorm`, ...

Exercise 2

Canadian-American astronomer and mathematician Newcomb conducted a series of experiments to determine the speed of light. The file `newcomb.RDS` on Canvas gives $n = 66$ measurements to be so interpreted: these

values times 10^{-3} plus 24.8 are the times, in millionths of a second, that light took to travel a known distance. *Note:* you do not need to manipulate the given values and can use them as they are for the exercise.

- a. Split the data into the first 20 and the last 46 observations, and consider these as two distinct samples. Explore the distribution of the data in the two samples graphically: for each, plot an histogram (scaled to density), a boxplot and the empirical CDF of each sample side-by-side, and comment on what you see; e.g. do the samples look like they could originate from a normal distribution, is the data distribution symmetric, are there any outliers...
- b. Assuming the first 20 observations originate from distribution F and the last 46 from distribution G , you want to test $H_0 : F = G$ vs. $H_1 : F \neq G$. Given what you have observed in a., which test (Wilcoxon two-sample test or Kolmogorov-Smirnov two-sample test) do you deem more suitable for this task and why?
- c. Perform the test you chose in b. and comment on the results (use a significance level of $\alpha = 5\%$).

Deliverables

- For all subtasks, to aid in correction: all relevant code
- The graphical output and comment required by a.
- The decision and corresponding explanation for b.
- The test results and comment for c.

Useful R functions

`readRDS`, `hist`, `boxplot`, `ecdf`, `plot`, `ks.test`, ...